

Total No. of Questions : 6]

SEAT No. :

P5115

[Total No. of Pages : 2

B.E./Insem. - 522

B.E. (Mechanical Engineering) (Semester - I)

CAD/CAM AND AUTOMATION

(2012 Pattern)

Time :1 hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Figures to the right side indicate full marks.
- 3) Use of Electronic pocket calculator is allowed.
- 4) Assume suitable data, if necessary.

Q1) a) A line is drawn with vertices A (5, 5) Q (10, 10) has undergone following transformation in sequence. **[6]**

- i) Rotation through 45° clockwise.
- ii) Reflection about line y axis.

Find the concatenated matrix and new coordinate of line.

b) Mention the steps to scale entity about any reference point P (xr, yr) with sketeches and related concatenated matrix. **[4]**

OR

Q2) a) Explain orthographic Projections for FV, TV and RSV with sketch and matrix. **[6]**

b) Triangle is defined by points P (2,2) Q (6,2) R (4,4) is created in User coordinate system (UCS) having x axis oriented a 45° and located at 4,5 units from Fixed Model coordinate system (MCS) Find coordinate of triangles in MCS. **[4]**

Q3) a) Compare analytical and parametric representation of curves with example. **[4]**

b) Write equation of line having P1 [3 5 8] and P2 [6, 4, 3]

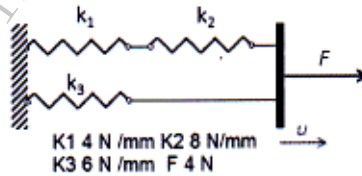
And find tangent vector and points on line at parameter $\mu = 0, 0.25, 0.5, 0.75, 1$ **[6]**

P.T.O.

OR

- Q4)** a) Explain Hermite Cubic Spline curve with mathematical formulation. [6]
b) Explain B - Rep and CSG methods of solid modeling. [4]

- Q5)** a) Explain the concept of shape function for 1 D elements. [4]
b) Cluster of springs are shown in figure calculate displacement and reaction force. [6]



OR

- Q6)** a) Find displacement, Stresses and reaction force for bar element shown in figure. [10]

